ChatGPT

Statistically Defensible Repeated Measurement for Live Browser Performance

Measurement target and first principles

The central methodological issue is not “which summary statistic should replace the arithmetic mean?” It is **what stochastic quantity the profiler is intended to characterize**.

For a fixed URL and nominal browser-agent profile, a measured performance outcome such as page-load time, Largest Contentful Paint, time-to-interactive, or another latency metric should be regarded as a realization of a random variable

$$Y \sim F(\cdot \mid C),$$

where C is the complete measurement condition: URL/version, browser and browser version, hardware, operating system, cache state, connection reuse, DNS state, network path, geographic location, browser profile, server/CDN state, time period, and measurement protocol. Systems research has repeatedly shown that apparently innocuous experimental conditions can materially change measured performance, and modern timing measurements may be neither normally distributed nor independent and identically distributed. [1]

This makes the **sampling protocol part of the estimand**. A profiler that reloads a URL immediately in the same browser session is not sampling the same population as one that launches a fresh browser, clears caches, obtains a new network connection, or waits 20 minutes between measurements. WProf, for example, explicitly found materially different page-load distributions for cold, warm, and immediately repeated “hot” loads, while Vesper found systematic differences between cold/warm cache conditions and between browsers. [2]

Consequently, the most defensible conceptual target is not initially a mean, median, or p95. It is the **conditional run-performance distribution** F_C, or equivalently its quantile function

$$Q_C(p) = F_C^{-1}(p),$$

for a precisely declared condition C. Frachtenberg and colleagues make essentially this distribution-first argument for contemporary computer-performance evaluation: increasingly complex systems produce performance measurements for which a single point summary can obscure important structure. [3]

For the proposed profiler, I would therefore separate four statistical objects:

Object	What is being estimated	Why it matters
Performance distribution	$F_C(y)$, represented empirically by an ECDF or quantile curve	Preserves skew, tails, mixtures, and changes in shape instead of forcing them into one number.
Operational location/tail summaries	One or more of $Q(.50), Q(.75), Q(.90), Q(.95)$, and possibly $E[Y]$	Answers specific questions such as “typical,” “slow for 1 in 10 runs,” or “expected latency.”
Dispersion/shape	IQR, MAD or another robust scale; perhaps SD/CV where meaningful; robust skewness	Distinguishes a consistently slow site from an erratic one with the same central value.
Uncertainty	Confidence interval or confidence sequence for each estimand	Expresses uncertainty caused by having only finitely many repeated measurements.

That architecture does **not** prejudge which percentile should ultimately be primary. It also avoids an important category error: a percentile is a characteristic of the performance distribution, whereas a confidence interval characterizes uncertainty about an estimated characteristic.

A further distinction should be built into the profiler from the beginning: a run can end in a **valid latency observation**, a **valid operational failure** such as a timeout that belongs to the target process, or a **measurement/protocol failure** caused by the profiler itself. Treating operational failures as merely “outliers” and deleting them would redefine the performance population after observing the data. A distribution-first approach instead motivates reporting failure probability separately where latency ceases to be meaningfully observed. This principle is consistent with systems work emphasizing that extreme timing observations and environmental effects can be part of genuine performance behavior rather than disposable noise. [4]

Likely distributional behavior and what must be tested empirically

There is good evidence against assuming a simple Gaussian, IID measurement model for the proposed problem. Chen and Revels explicitly report that empirical program-timing distributions are often heavy-tailed and that timing observations need not be IID because timer error, operating-system jitter, and environmental fluctuations can generate strongly non-ideal statistics. Mytkowicz et al. demonstrated substantial performance-measurement bias from seemingly innocuous changes in experimental conditions, while STABILIZER was motivated by the fact that low-level layout effects can make conventional repeated timing experiments difficult to interpret statistically. [5]

For live web pages there are additional stochastic mechanisms. WProf decomposed page loading into browser computation and network activity, finding browser computation responsible for about 35% of its measured critical path in its main experiments; external analytics/advertising resources were on the critical path for a substantial subset of pages; and cold, warm, and hot-cache page-load distributions differed materially. [6] Vesper likewise showed that cache state, browser choice, and network RTT changed measured performance, including different effects at the median and 95th percentile. [7]

Those results imply several distinct statistical possibilities for the profiler:

Positive skew is plausible but should not be imposed as a model assumption. Latencies are nonnegative, and occasional scheduling pauses, server delays, retransmissions, third-party delays, garbage collection, or cache misses naturally generate observations substantially larger than the main body of measurements. Empirical systems-timing research indeed reports heavy-tailed timing distributions. [8] A particular tightly controlled URL might nevertheless produce a nearly symmetric or extremely concentrated distribution, so “web performance is lognormal” or “web performance is heavy-tailed” would be too strong as a universal premise.

Heavy tails are especially important because they make moment-based summaries fragile. A relatively small number of legitimate long runs can move the mean, variance, SD, CV, and third-moment skewness substantially. Chen and Revels specifically identify this problem for timing measurements. [8] Dean and Barroso's tail-latency work provides a separate operational reason not to dismiss slow observations in distributed systems: rare slow components can matter disproportionately to end-to-end latency. [9] That does not itself justify p95 for browser tests, but it does justify preserving rather than automatically trimming the right tail.

Multimodality should be treated as a serious possibility, particularly when latent states are pooled. WProf's cold/warm/hot results show that cache state changes page-load behavior; therefore, pooling those states can mathematically create a mixture distribution even if each state separately is unimodal. [6] Similar mixtures can arise from alternative CDN/server paths or from third-party resources being present in only some runs. The latter mechanisms are plausible explanations that should be diagnosed from profiler traces rather than presumed.

Heteroscedasticity is likely across experimental cells even when it is absent within a cell. Different sites, cache states, RTTs, browsers, and hardware conditions need not merely shift the center; Vesper demonstrates that effects can differ between the median and p95, indicating changes in distributional shape rather than a simple common additive offset. [7] Kalibera and Jones similarly emphasize identifying the levels at which performance variance arises when designing benchmark repetitions. [10]

Temporal dependence is a first-class concern. An immediately repeated load can inherit browser cache entries, DNS entries, persistent connections, JIT state, process state, CDN state, or server/network conditions from preceding measurements. WProf's definition of a hot load as a load immediately following another is an explicit example in which run order changes the experimental state. [6] Chen and Revels independently warn that timing measurements may be non-IID. [8]

The empirical diagnostic phase should therefore preserve **run order** and record all available state metadata. For every URL/profile/condition, examine the ECDF, histogram or density estimate, quantile plot, run-order plot, and lag structure; additionally stratify by cache/session state and major categorical events such as navigation failure or third-party failure. Distribution plots should be inspected on the original and, for positive timing variables, log scale. A statistical normality test can supplement this analysis, but the framework should not depend on passing a normality test before selecting a method; systems-performance literature provides ample reason to use inference that remains useful outside the Gaussian model. [11]

A useful evidence classification is therefore:

Property	Evidence for treating it as plausible	Position for this research
Exact normality	Weak; timing research explicitly reports non-ideal/heavy-tailed behavior. [8]	**Do not assume.**
Positive/right skew	Mechanistically plausible and consistent with timing-tail evidence. [12]	**Expect often, verify per condition.**
Heavy tails	Directly reported in systems timing literature. [8]	**Design inference to tolerate them.**
Multimodality	Strongly plausible when cache or other states are mixed; WProf demonstrates distinct cache-state distributions. [6]	**Diagnose mixtures before summarizing.**
Heteroscedasticity	Different web/browser conditions alter more than just central location. [13]	**Allow group-specific scale.**
Serial dependence	Timing need not be IID; sequential loads explicitly change state. [14]	**Test and design for it.**
Stationarity over long tests	Live server/CDN/network environment can vary; distribution-first systems work warns against assuming a simple fixed distribution. [3]	**Check drift/change points; time-block experiments.**

The important limitation is that **no published benchmark can establish the exact distribution of your profiler**. The literature justifies a robust measurement design and a prespecified diagnostic study. The profiler's own large-repetition pilot data must establish whether these phenomena occur under its particular URL/profile/reset protocol.

Estimands, variability, shape, and uncertainty

An estimand should be selected because it corresponds to an operational question, not because it is statistically fashionable.

For a latency variable where *smaller is better*, the population mean,

$$\mu = E[Y],$$

answers: **"What latency would a randomly sampled run contribute on average under this sampling protocol?"** The mean uses every magnitude, including rare extreme delays. That is an advantage when the cumulative cost of slow events matters, but a disadvantage when a small number of highly variable tail observations makes the estimate unstable. Systems timing data can have exactly this behavior. [8]

The median,

$$Q(.50),$$

answers: **"Half of runs are at least this fast and half are at least this slow."** Its interpretation remains straightforward under positive skew and it is much less sensitive than the mean to the magnitude of very

slow observations. It does not, however, answer anything about how bad the slow tail is. A site that is normally fast but occasionally catastrophically slow can have an excellent median.

A p75 answers: **"75% of runs are no slower than this value; 25% are slower."** There is a real operational precedent for p75 in web performance: Google's Core Web Vitals chose the 75th percentile to balance representing the majority of page visits against increasing sensitivity to outliers at higher percentiles. [15] That argument applies to a distribution of **real-user visits**, however, and is not by itself a statistical justification for p75 in a synthetic repeated-run experiment with one browser profile. The population and scientific question are different.

A p90 asks for a **one-in-ten slow-run boundary**, while p95 asks for a **one-in-twenty slow-run boundary**. Increasing p moves closer to the tail and can therefore make the measure more operationally sensitive to intermittent slowness, but the price is sharply increasing sampling uncertainty. There is no mathematical privilege attached to 0.95 rather than 0.90, 0.925, or 0.99. The percentile should correspond to a stated performance objective. Frachtenberg et al.'s distribution-first position is particularly relevant here: retaining the distribution avoids pretending that one percentile universally represents performance. [3]

The following comparison gives a more complete candidate set.

Class	Statistic	Operational interpretation	Strengths	Important weaknesses	Suitable uncertainty method
Location	**Mean**	Expected latency of a run sampled according to the protocol	Uses complete magnitude information; natural for expected total cost	Tail-sensitive; can converge slowly under strong skew/ heavy tails	Studentized interval when assumptions reasonable; bootstrap-t/ BCa otherwise; block bootstrap under dependence. [16]
Location	**Median / p50**	Typical central threshold	Robust to extreme magnitudes; interpretable under skew	Can completely hide slow-tail severity	Exact/ binomial order-statistic interval or bootstrap. [17]

Class	Statistic	Operational interpretation	Strengths	Important weaknesses	Suitable uncertainty method
Upper-central	**p75**	25% of runs are slower	More tail-sensitive than median; substantial web-field precedent	That precedent concerns field visits, not synthetic repetitions	Order-statistic or bootstrap interval. [18]
Tail	**p90**	10% of runs are slower	Useful compromise between tail sensitivity and sample demand	Still ignores severity among slowest 10%; more uncertain than p75	Order-statistic interval; bootstrap with adequate n. [17]
Tail	**p95**	5% of runs are slower	Direct tail/SLO interpretation; detects intermittent performance not reflected in median	Extremely data-hungry at small n; top few observations determine estimate	Exact order-statistic CI preferred for transparent finite-sample inference; quantile confidence sequence for valid IID sequential monitoring. [19]
Spread	**Variance / SD**	Mean-squared or same-unit dispersion around mean	Familiar; useful with mean and Gaussian-like distributions	Strongly driven by extreme observations and mixtures	Parametric interval only if model defensible; otherwise bootstrap. [20]
Relative spread	**CV = SD/ mean**	Dispersion relative to mean	Dimensionless; facilitates scale-normalized comparisons	Inherits sensitivity of mean and SD; interpretation becomes awkward in strongly skewed distributions	Bootstrap or specialized CV methods; do not use normal-theory formulas blindly. [21]

Class	Statistic	Operational interpretation	Strengths	Important weaknesses	Suitable uncertainty method
Robust spread	**IQR = p75−p25**	Width of middle 50%	Simple, robust, same units as measurement	Insensitive to tail instability by design	Order-statistic/ bootstrap interval
Robust spread	**MAD**	Typical absolute deviation from median	Highly robust, same units after optional consistency scaling	Less directly interpretable to product users than IQR	Bootstrap; S_n and Q_n are robust alternatives. [22]
Relative robust spread	**IQR/median or MAD/ median**	Scale-normalized robust variability	More resistant than CV to extreme timing observations	Still a ratio; requires consistent definition and positive median	Robust relative-dispersion intervals or bootstrap. [21]
Shape	**Moment skewness**	Standardized third-moment asymmetry	Familiar	Highly tail-sensitive; unsuitable as sole shape diagnostic in contaminated/ heavy-tailed data	Bootstrap if used
Robust shape	**Medcouple / quantile skewness**	Robust distributional asymmetry	Much less outlier-sensitive; medcouple has bounded influence and 25% breakdown in its original analysis	Less familiar to practitioners	Bootstrap; descriptive use often sufficient. [23]
Uncertainty	**Confidence interval**	Uncertainty about any one of the above population quantities	Separates point estimate from inferential precision	Not itself a performance percentile	Method depends on the estimand and dependence structure

A particularly important warning concerns the **coefficient of variation**. Arachchige, Prendergast, and Staudte specifically argue that because CV is based on the mean and SD, outliers adversely affect it and interpretation can be difficult for skewed distributions; they study IQR- and MAD-based relative-dispersion alternatives and their interval estimation. [24] Therefore CV is reasonable as a supplementary metric when the run distribution is reasonably homogeneous and mean-centered interpretation makes sense, but it should not become the profiler's universal definition of “stability.”

Similarly, classical third-moment skewness is not an ideal headline measure for a tail-prone timing distribution. The medcouple was explicitly developed as a robust alternative to classical skewness, with bounded influence and a 25% breakdown value. [23] In practice, an ECDF plus p50/p75/p90/p95 and a robust spread conveys more operational information than reporting a skewness coefficient alone.

There is also an easily overlooked reproducibility issue: **“the p95” does not completely specify the finite-sample estimator.** Hyndman and Fan documented multiple definitions of sample quantiles used by statistical software; interpolation rules can differ substantially for small samples. [25] The profiler should therefore explicitly specify the quantile algorithm used. Interpolating between the two largest observations in a 20-run experiment may produce a numerically smooth “p95,” but it does not create information about an unobserved population tail.

Percentiles are not confidence intervals

Suppose the profiler reports

$$\widehat{Q}(.95) = 4.2\ \mathrm{s}, \qquad 95\%\ \mathrm{CI} = [3.7, 5.4]\ \mathrm{s}.$$

The **4.2 s p95** means that the estimated latency boundary has 95% of the target run distribution below it.

The **95% confidence interval** refers instead to repeated-sampling coverage of the unknown population p95 by the interval-building procedure. Its “95” has no connection to the percentile's “95.” One could legitimately report a 90% CI for p95, a 99% CI for p75, or a 95% CI for the mean. Nonparametric quantile-interval literature explicitly treats the confidence level and target quantile as separate parameters. [17]

For candidate CI methods:

Means. Under IID normal data the classical Student-t interval has its usual exact justification; under sufficiently regular finite-variance nonnormal distributions it has asymptotic justification. Systems benchmarking papers such as Georges et al. have used mean confidence intervals to determine measurement precision, but browser timing's potential skew/heavy tails make the assumptions something to investigate rather than presume. [26] Bootstrap-t or BCa intervals are useful nonparametric alternatives when IID sampling is credible; Efron's BCa construction was designed to improve bootstrap treatment of bias and skewness. [27]

Medians and other quantiles. Under IID continuous sampling, an exact distribution-free interval can be obtained from order statistics because the number of observations below the true p-quantile has a binomial distribution. Hutson develops nonparametric confidence intervals for arbitrary quantiles, including median and quartiles. [17] These intervals are especially attractive because they make the limited information about a p95 at small n completely visible.

Relative variability. For skewed timing data, bootstrap intervals for CV or robust ratios are more defensible than blindly importing normal-theory CV inference. Arachchige et al. provide specific interval methods for robust IQR/MAD relative-dispersion alternatives. [21]

Dependent observations. An ordinary IID bootstrap is not appropriate merely because it is "nonparametric": it destroys temporal structure by resampling individual observations independently. The stationary bootstrap of Politis and Romano was explicitly developed to resample stationary weakly dependent observations for statistics such as standard errors and confidence intervals. [28] Block/bootstrap methods should therefore replace an IID bootstrap when run-order dependence remains after experimental controls.

Required repetitions and why high percentiles are difficult

There is no defensible universal statement such as "run every URL 10 times," "30 repetitions are enough," or even "100 repetitions are enough." Systems benchmarking work has repeatedly found that required replication differs across benchmarks and experimental levels. Georges et al.'s Java evaluation showed materially different confidence-interval precision across workloads even at substantial repetition counts, and their JavaStats approach continued measurements until a requested precision or a maximum invocation count was reached. [29] Kalibera and Jones similarly argue that repetitions should be allocated according to variance arising at different experimental levels rather than uniformly by convention. [10]

For a **mean**, an approximate IID planning relationship is

$$n \approx \left(\frac{z_{1-\alpha/2}\sigma}{h} \right)^2,$$

where h is the desired CI half-width. A pilot estimate of σ can replace the unknown population value for planning, but this approximation can be poor when the pilot misses a heavy tail or when observations are dependent. The systems evidence against casually assuming IID, light-tailed timings is therefore directly relevant. [8]

For a **quantile**, sample size depends not only on p but on how densely probability is concentrated near the target quantile. The familiar large-sample form

$$\mathrm{SE}(\widehat{Q}(p)) \approx \frac{\sqrt{p(1-p)}}{\sqrt{n}\, f(Q(p))}$$

shows the difficulty: if the density f is low in the upper tail, a small change in percentile rank corresponds to a large change in milliseconds. Thus there cannot be a universal n guaranteeing, for example, a ±5% relative CI around p95 for all websites. Nonparametric order-statistic intervals avoid estimating f, but their width in latency units will likewise reveal that uncertainty. [17]

A distribution-free lower bound on p95 information

The p95 problem can be demonstrated without assuming a lognormal, gamma, or any other latency distribution.

For IID observations from a continuous distribution, let $q_{.95}$ denote the true p95. The probability that the sample maximum exceeds that true p95 is

$$P(X_{(n)} \geq q_{.95}) = 1 - 0.95^n.$$

Equivalently, the expected number of observations beyond the true p95 is only

$$n(1 - .95) = 0.05n.$$

The following values follow directly from the binomial/order-statistic construction used for distribution-free quantile inference. [17]

Runs n	Expected observations in slowest 5%	Probability sample maximum reaches/ exceeds true p95	Consequence
10	0.5	40.1%	The p95 is effectively an extreme-order extrapolation/interpolation problem.
20	1.0	64.2%	Very little tail information.
30	1.5	78.5%	Still too sparse for a well-resolved p95.
50	2.5	92.3%	Better, but not enough even for a 97.5% one-sided upper order-statistic bound.
59	3.0	95.2%	Approximately the minimum for the sample maximum to give 95% one-sided coverage of the p95.
72	3.6	97.5%	Minimum at which the maximum can satisfy the upper side of an equal-tail 95% distribution-free interval.
100	5.0	99.4%	Only five expected observations lie beyond the population p95.
200	10.0	>99.99%	Far better tail resolution, but latency-unit CI width can still be large if the tail is sparse.

The $n = 59$ and $n = 72$ thresholds are **not recommended sample sizes**. They are much weaker: they are mathematical floors for obtaining particular finite upper order-statistic coverage properties. At p95, even 100 runs contain only five expected observations beyond the target quantile. Dependence between sequential runs reduces the effective independent information further, so these IID calculations are optimistic when positive autocorrelation exists. [30]

The same calculation illustrates how rapidly the difficulty increases with percentile level. To make the sample maximum exceed the true target quantile with at least 97.5% probability requires approximately:

Target quantile	Minimum IID n satisfying $p^n \leq 0.025$
p75	13
p90	36
p95	72
p99	368

Again, these values guarantee nothing about a **narrow** CI. They only show when a finite sample maximum can provide the requisite upper-tail coverage. This is one of the strongest reasons not to select p95 or p99 simply because those percentiles are common in service dashboards. Exact nonparametric quantile intervals make this sample-information limitation explicit. [17]

This also gives a statistical argument for considering p90 when the repetition budget is constrained. A p90 has twice as much probability mass above it as p95, so it supplies roughly twice as many tail observations at a given n. Whether that gain in precision is worth giving up the one-in-twenty interpretation is an **operational** decision, not one statistics can resolve by itself.

A sound sample-size procedure should therefore start from a required precision:

$$\text{absolute width} \leq \delta$$

or, for a positive estimand,

$$\text{relative width} = \frac{U - L}{|\hat{\theta}|} \leq r,$$

with the exact convention—full width versus half-width—specified in advance. Required n should then be estimated from a reasonably large pilot or obtained adaptively. This reverses the conventional workflow: do not select n first and accept whatever precision occurs; select scientifically useful precision and determine the measurement burden required to obtain it. Georges et al. and the later adaptive-performance literature both motivate this precision-driven direction. [31]

Sequential and adaptive stopping

Adaptive stopping is attractive for this profiler because easy, stable URLs should not consume the same measurement budget as highly variable URLs. But there is a major inferential trap: **repeatedly recomputing an ordinary fixed-sample 95% CI and stopping the first time it becomes narrow is not automatically a valid 95%-coverage procedure.**

Singham and Schruben investigated precisely this issue for absolute-precision stopping rules. They found that a basic CI-width stopping rule can produce biased finite-sample coverage **even when the underlying observations satisfy the procedure's IID normal assumptions**. Their work develops coverage-adjusted stopping rules rather than treating “CI is finally narrow enough” as inferentially innocuous. [32]

That leads to three increasingly sophisticated approaches for this application.

A two-stage or batched precision design is the simplest defensible implementation. Collect a prespecified pilot n_0, use it to estimate the scale or quantile uncertainty, determine an additional sample requirement from the declared precision target, and then collect that additional sample before making final inference. The important advantage is that the final CI need not be peeked at after every observation. Benchmarking work by Georges et al. and Kalibera and Jones provides the systems-performance motivation for determining repetition from estimated uncertainty rather than a universal fixed count. [33]

A properly calibrated fixed-width sequential procedure is the classical sequential solution. Rather than applying ordinary CIs after every sample, the stopping rule itself is constructed to maintain coverage under its assumptions. Singham and Schruben show why finite-sample calibration matters. [32] Such procedures can work well when their distributional and independence assumptions are plausible, but browser measurements make those assumptions an empirical question.

Confidence sequences are the cleanest modern solution when true continuous monitoring is required. A confidence sequence is designed to maintain its stated coverage simultaneously over time, which means a data-dependent stopping time does not invalidate the interval merely because the experiment stopped when the desired precision appeared. Howard, Ramdas, McAuliffe, and Sekhon developed nonparametric, nonasymptotic time-uniform confidence sequences for several parameters under appropriate stochastic conditions. [34] More directly relevant, Howard and Ramdas developed confidence sequences for fixed quantiles—and even simultaneous tracking of all quantiles—from streams of IID observations. [35]

The last qualification is critical: the Howard–Ramdas quantile method assumes an IID stream. It should not simply be pasted onto sequential browser runs with persistent-cache and server-state dependence. [36] If the measurement protocol can genuinely produce approximately independent fresh-state runs, a quantile confidence sequence is exceptionally well aligned with the proposed profiler. If dependence remains intrinsic, a conservative batched design plus block/bootstrap treatment of dependence is easier to defend statistically. [28]

There is also a newer systems-specific approach. Mittal and colleagues' **Adaptive Stopping Rule for Performance Measurements** treats the performance distribution as the primary measurement target and proposes a meta-heuristic that dynamically determines when the collected sample sufficiently approximates that distribution; they report advantages over fixed stopping criteria and explicitly avoid requiring an advance parametric form for the performance distribution. [37] This is closely aligned conceptually with the browser profiler, but it should be described as an adaptive distribution-characterization method rather than assumed to provide the same finite-sample inferential guarantee as an anytime-valid confidence sequence.

A practical profiler protocol could therefore use this logic:

$$\boxed{\text{minimum information} \rightarrow \text{state/dependence checks} \rightarrow \text{precision assessment} \rightarrow \text{continue or stop}}$$

rather than

$$\boxed{\text{run 5 or 10 times and average.}}$$

The stopping criterion should be attached to the **chosen estimand**. For a median it could be the width of its nonparametric CI; for p90/p95, the quantile CI; for an entire distribution, a distributional discrepancy/band criterion; for a relative-variability parameter, the CI for that parameter. Merely checking that the newest point estimate moved by less than, say, 1% is much weaker: a statistic can appear numerically stable before its sampling uncertainty is acceptably small. The distribution-focused systems literature specifically argues for stopping based on characterization of the distribution rather than arbitrary repetition counts. [38]

A strong engineering specification would therefore contain a **minimum** n, a **maximum** n, a declared **precision requirement**, and a declared response when precision cannot be achieved before the maximum. The last case should be reported as "high uncertainty/unstable distribution," not silently forced into the same-quality result as a stable URL. This is precisely the kind of heterogeneity in measurement demand seen in rigorous benchmarking studies. [33]

For p95 in particular, I would not let an adaptive rule stop at very small n merely because an interpolated p95 happens not to have moved recently. The order-statistic calculations above show why: at $n = 20$ the population tail has barely been sampled. A defensible p95 rule should have a substantial p95-specific minimum sample floor **in addition to** a precision criterion. [39]

Experimental protocol: warm-up, cache state, outliers, dependence, and hierarchy

Warm-up should be defined scientifically, not discarded mechanically. The common benchmarking practice of dropping early observations under the assumption that they precede a stable warmed-up regime is not universally justified. Barrett et al.'s large VM warm-up study used changepoint analysis and found highly varied warm-up behavior; at most 43.5% of the VM/benchmark combinations in their experiments consistently reached a steady peak-performance state across the studied machines. [40]

For a browser profiler the argument is stronger still because "cold" may be an operational state of genuine interest rather than contamination. WProf deliberately measured cold, warm, and hot page loads and found large differences; caching reduced page-load time by more than 40% for half of the pages in its experiments, while also affecting both networking and browser computation. [6] A first cold navigation should therefore not be deleted merely because it is slower.

The right protocol depends on the desired estimand:

- **Fresh-visit performance:** reset browser/cache/network state according to an explicit definition before each measured run.
- **Warm repeat-visit performance:** establish a reproducible warm-state protocol and measure that distribution.
- **Natural sequential browsing/reloading:** preserve the dependence because it is part of the process being studied.
- **Both cold and warm behavior:** treat them as separate experimental strata and estimate separate distributions.

These distinctions follow directly from the empirical cache-state differences documented by WProf and Vesper. [2]

Outlier decisions should be based on provenance, not distance from the median. A 10-second run amid 2-second runs is not a measurement error merely because a boxplot labels it an outlier. It might be the exact third-party stall, scheduler delay, server queue, or network event that a tail statistic is meant to represent. Systems timing literature explicitly treats environmental variability and heavy-tailed timing observations as genuine features that robust benchmarking must cope with. [12]

A defensible exclusion rule is therefore procedural: exclude a run only when there is independent evidence that it violated the measurement protocol—for example, the profiler failed to start instrumentation, the browser crashed because of the harness, the requested URL was not actually the target, or the instrumentation record was corrupt. Record the exclusion reason and retain the raw record in the audit trail. Mytkowicz et al.'s demonstration that subtle setup choices can bias systems measurements is a strong argument for this provenance-based discipline. [41]

Timeouts deserve special treatment. When a timeout represents a genuine outcome possible under the defined live-URL environment, deleting it would systematically remove the worst behavior. A particularly transparent design is to estimate **failure/timeout probability** as one parameter and the latency distribution of successfully completed runs as another, while preserving timeout information for any tail analysis. This is more faithful to a distribution-first performance model than replacing failures with an arbitrary fabricated latency or deleting them. [42]

Dependence should be attacked both experimentally and analytically. Where the scientific target is an independent fresh navigation, reset state, space or randomize runs when appropriate, and interleave competing browser profiles over time rather than completing all measurements for profile A and then all of profile B. Systems methodology such as Duet benchmarking shows the value of designs that expose competing configurations to common time-varying environmental noise rather than confounding treatment with time. [43] Where dependence remains, preserve run order and use a dependence-aware inferential method such as a block or stationary bootstrap rather than treating all observations as exchangeable. [28]

Hierarchical structure should be modeled rather than flattened. A realistic research dataset might contain observations indexed as

$$Y_{\text{website, profile, day/session, run}}.$$

Repeated runs within the same session are not equivalent to independent websites; likewise, measurements from many runs on one website do not magically create a large sample of websites. Kalibera and Jones explicitly model systems benchmarking as having multiple levels of repetition and recommend locating where variance enters the experiment so measurement effort can be allocated intelligently. [10]

For a multi-website/profile study, a mixed/hierarchical model could include website effects, profile effects, website×profile interactions, session/day effects, and run-level residual variation, potentially with group-specific scales or a dependence model for sequential residuals. Whether “website” or “profile” should be a random effect is a design question: if the exact tested websites are the population of interest, fixed effects are appropriate; if websites were genuinely sampled to generalize to a broader population, a population-level random-effects interpretation becomes meaningful. The central methodological point is that **the independent experimental unit must not be confused with the number of repeated measurements**.

Kalibera and Jones's hierarchical treatment of benchmark repetitions provides the closest systems-specific methodological precedent. [10]

For highly non-Gaussian data, the hierarchical analysis does not have to force raw latency into a normal random-effects model. Options include a transformation such as log latency when scientifically appropriate, a parametric positive-valued model only after diagnostics, hierarchical bootstrap at independent cluster levels, or quantile-oriented modeling if the research question itself concerns conditional medians/tails. The trade-off is that more elaborate models gain efficiency and cross-site inference by imposing additional structure; the distribution-free within-condition approach makes fewer assumptions but requires more observations.

Candidate frameworks and overall assessment of p95

The evidence favors several legitimate frameworks rather than one universally correct statistic.

Distribution-first robust framework

The fundamental estimand is F_C, represented by the ECDF or quantile function for each controlled browser condition. Report a compact quantile set—at minimum p50 plus one or more upper quantiles justified by operational goals—together with IQR or MAD and confidence intervals. Add robust relative dispersion, such as IQR/median or MAD/median, when cross-site scale-normalized variability is useful. This follows the recent distribution-first systems argument while using well-established nonparametric quantile and robust-scale methodology. [44]

Its main strengths are few distributional assumptions, resistance to isolated extreme magnitudes, visibility of tail behavior, and straightforward operational interpretation. Its weaknesses are increased reporting complexity and substantial sample demands for high quantiles.

This is the strongest general-purpose starting framework for the profiler.

Estimand-specific precision framework

Choose a specific primary estimand because the product/research objective demands it—mean, median, p75, p90, or p95—and continue measurement until the corresponding inferential precision is adequate. Use exact/nonparametric quantile intervals for quantiles, appropriate mean intervals for means, and dependence-aware methods where necessary. Systems benchmarking already contains precedent for precision-driven measurement rather than arbitrary counts, but classical stopping-rule research warns that naïvely peeking at ordinary CIs can invalidate coverage. [45]

This is attractive when a single number is needed for automated comparison or regression detection. Its weakness is that the chosen estimand can miss important changes elsewhere in the distribution.

Sequential distribution framework

Use a minimum sample and then a sequentially valid method. For IID conditions, confidence sequences—particularly Howard and Ramdas's quantile confidence sequences—provide a principled way to stop at a

data-dependent time while retaining time-uniform coverage. [46] A systems-specific alternative is the adaptive distribution-characterization heuristic proposed by Mittal et al. [37]

This is potentially the most computationally efficient architecture, but it has the greatest implementation burden. Quantile confidence sequences have assumptions, especially IID sampling, which a live browser test may violate unless the experimental state is aggressively controlled. [36]

Hierarchical experimental framework

For research across multiple websites, profiles, days, and sessions, estimate within-condition distributions while simultaneously modeling higher-level sources of variability. This distinguishes repeatability of one URL/profile from generalizability across websites and profiles and follows the multi-level repetition logic of rigorous systems benchmarking. [10]

Its weakness is model dependence and substantially greater analytical complexity. It is complementary to, rather than a replacement for, careful within-condition distributional measurement.

Is p95 justified?

p95 has a legitimate operational interpretation but no unique statistical justification. It is appropriate when the research question genuinely concerns a one-in-twenty slow-run boundary—for example, intermittent tail degradation that a median or p75 would deliberately hide. Tail latency is important in computer systems, so preserving such a tail estimand can be scientifically meaningful. [47]

However, the statistical cost is substantial. With 20 independent runs there is only one expected observation in the population's slowest 5%; with 100 there are five. Exact distribution-free quantile inference shows that fewer than 72 IID observations cannot even give the sample maximum the upper-side coverage needed for an equal-tail 95% p95 interval, let alone guarantee a useful narrow interval. [17] Positive temporal dependence makes these IID figures optimistic. [30]

Therefore **p95 is weak as the profiler's sole universal headline statistic**, especially if measurement budgets are in the conventional 5–30-run range. It becomes much more defensible if tail performance is genuinely an objective, the profiler collects substantially larger/adaptive samples, quantile uncertainty is reported explicitly, and sequential dependence is controlled or modeled.

p90 is statistically easier to estimate for a fixed measurement budget while still representing a clearly tail-oriented “one in ten runs slower” boundary. p75 is substantially more stable and has a field-web precedent, but Google's rationale concerns distributions of real-user experiences rather than synthetic repetitions and should not be imported uncritically. [15] Median is an excellent typical-performance estimand but insufficient alone whenever intermittent slowness matters.

The most defensible architecture at this stage of the research is therefore **not to choose between mean, median, p75, p90, and p95 prematurely**. A large-repetition pilot should estimate the distribution and determine how precision degrades as the requested percentile moves into the tail. The eventual selection can then be based on both operational meaning and attainable uncertainty.

A particularly strong candidate reporting model would be:

$\mathrm{ECDF/quantile\ distribution} + \mathrm{p50} + \mathrm{predeclared\ upper\ quantile} + \mathrm{IQR\ or\ MAD} + \mathrm{CI\ for\ each\ inferential\ estiman}$

with p95 added or promoted only when its operational role and measurement precision support doing so. This matches the distribution-oriented direction of recent performance-methodology research without assuming in advance that any one conventional statistic is "the" performance measure. [38]

Priority methodological papers to obtain

The following literature forms a strong methodological reading set for the proposed research. The first group is particularly important because it is directly about rigorous computer-systems performance measurement rather than generic statistics.

Priority paper	Why it matters
Kalibera & Jones, "Rigorous Benchmarking in Reasonable Time," ISMM 2013	One of the most directly relevant systems-methodology papers: hierarchical repetition levels, variance decomposition, repetition allocation, and uncertainty-aware benchmarking. [48]
Georges, Buytaert & Eeckhout, "Statistically Rigorous Java Performance Evaluation," OOPSLA 2007	Foundational practical work on repeated performance measurements, confidence intervals, non-determinism, and adaptive measurement based on desired CI width. [49]
Mytkowicz, Diwan, Hauswirth & Sweeney, "Producing Wrong Data Without Doing Anything Obviously Wrong!," ASPLOS 2009	Essential experimental-design warning: apparently innocuous setup choices can systematically bias systems-performance conclusions. [41]
Curtsinger & Berger, "STABILIZER: Statistically Sound Performance Evaluation," ASPLOS 2013	Demonstrates how uncontrolled systems-level effects can frustrate conventional inference and uses randomization to improve statistical behavior. [50]
Chen & Revels, "Robust Benchmarking in Noisy Environments," HPEC 2016	Especially relevant to your distributional question: discusses non-IID measurements, heavy-tailed empirical timings, environmental noise, and robust benchmarking. [8]
Barrett et al., "Virtual Machine Warmup Blows Hot and Cold," OOPSLA 2017	Probably the most important paper for challenging arbitrary warm-up discards; uses changepoint analysis and shows that simple steady-state assumptions frequently fail. [40]
Frachtenberg et al., "The Distribution Is the Performance," IEEE Computer 2024	Direct argument that performance evaluation should increasingly characterize distributions rather than relying only on simple point summaries. [3]

Priority paper	Why it matters
Mittal et al., “Adaptive Stopping Rule for Performance Measurements,” PMBS/SC 2023	Highly relevant recent systems-specific work on stopping when the performance distribution has been sufficiently characterized rather than using a fixed repetition count. [51]
Singham & Schruben, “Finite-Sample Performance of Absolute Precision Stopping Rules,” INFORMS Journal on Computing 2012	Crucial corrective to naïve “stop when CI width is small enough”: demonstrates finite-sample coverage bias and studies properly calibrated stopping. [32]
Howard & Ramdas, “Sequential Estimation of Quantiles with Applications to A/B Testing and Best-Arm Identification,” Bernoulli 2022	Direct theoretical basis for anytime-valid sequential quantile estimation under IID sampling. Highly relevant if adaptive p75/p90/p95 stopping is implemented. [35]
Howard, Ramdas, McAuliffe & Sekhon, “Time-Uniform, Nonparametric, Nonasymptotic Confidence Sequences,” Annals of Statistics 2021	General theoretical foundation for valid continuously monitored inference and data-dependent stopping. [52]
Wang et al., “Demystifying Page Load Performance with WProf,” NSDI 2013	Direct browser evidence on computation/network contributions and, importantly, distinct cold/warm/hot-cache performance distributions. [6]
Netravali et al., “Vesper: Measuring Time-to-Interactivity for Web Pages,” NSDI 2018	Browser-specific empirical evidence that cache, browser, network RTT, median and p95 behavior differ; useful evidence that distribution shape matters. [7]
Hyndman & Fan, “Sample Quantiles in Statistical Packages,” The American Statistician 1996	Essential before implementing percentiles: establishes that multiple sample-quantile definitions exist and must be specified for reproducibility. [25]
Hutson, “Calculating Nonparametric Confidence Intervals for Quantiles Using Fractional Order Statistics,” Journal of Applied Statistics 1999	Direct methodology for distribution-free confidence intervals around medians and arbitrary quantiles; especially relevant for p90/p95. [17]
Efron, “Better Bootstrap Confidence Intervals,” JASA 1987	Foundational BCa bootstrap work; relevant to skewed estimators where ordinary normal intervals are unsatisfactory. [27]
Politis & Romano, “The Stationary Bootstrap,” JASA 1994	Foundational resampling method for stationary weakly dependent observations; directly relevant if sequential browser runs remain temporally correlated. [28]
Rousseeuw & Croux, “Alternatives to the Median Absolute Deviation,” JASA 1993	Introduces S_n and Q_n, useful robust scale alternatives when SD is too tail-sensitive. [22]

Priority paper	Why it matters
Brys, Hubert & Struyf, "A Robust Measure of Skewness," JCGS 2004	Primary medcouple paper; provides a much more robust shape measure than classical moment skewness. [23]
Arachchige, Prendergast & Staudte, "Robust Analogs to the Coefficient of Variation," Journal of Applied Statistics 2022	Directly relevant to the proposed use of CV; develops IQR/MAD-based relative-dispersion alternatives and interval estimators. [24]
Dean & Barroso, "The Tail at Scale," Communications of the ACM 2013	Important operational context for why latency tails can matter; valuable justification for studying upper-tail behavior, though not by itself a reason to select exactly p95. [53]

Taken together, this literature supports a methodological direction quite different from "five runs and an arithmetic average": **define the stochastic population first; preserve and diagnose the run distribution; distinguish genuine experimental states; quantify uncertainty for the estimand actually being reported; make tail sample size explicit; and, if using adaptive measurement, use a stopping procedure whose inferential properties survive the stopping rule itself.** [54]

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